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Design engineer building expressive interfaces from early ideas to shipped software.

University of Pennsylvania · BSE Digital Media Design (Computer Graphics) + MSE Computer Science · Expected 2027
Frontend Software Engineering Intern at PayPal · U.S. Permanent Resident · Available full time beginning August 2026

PROJECTS

Digital Loom — GenAI, Three.js, GLSL, TypeScript | 2026

- Material editor that turns a fabric photograph into tunable PBR textures, simulated cloth, and an exportable 3D asset.
- Developed AI-assisted map-extraction pipeline, custom warp/weft solver, parallax occlusion, procedural edge fraying.

Into the Blue: Virtual Archaeology Museum Experience — React, Next.js | 2025

- Proposed camera-to-sticker interaction for virtual artifact hunt, then built the browser-based capture, cutout, and offline-storage pipeline with a five-person team.
- Deployed museum-wide for nearly one year at the Penn Museum, which serves 180K+ visitors annually.

Internet Atlas — Three.js, D3.js, React, Pinecone, FastAPI, Supabase | 2025

- Co-led product and technical direction for a 3D browsing-history map of 300+ websites and 1,000+ connections.
- Connected the team's ML pipeline to an interactive graph organized by visitor-chosen words; built 3D and 2D interactions, data model, backend integration, and deployment.
- Supported 40 simultaneous users during a live demo on \$5 hosting budget, then converted site into static deployment

Mini Minecraft — C++, OpenGL, GLSL, Qt | 2024

- Built a voxel engine from scratch with a three-person team, including multithreaded chunking, procedural terrain, sky shaders, and real-time water reflections.

Interaction & Graphics Experiments (live at estellekimdev.com)

WORK EXPERIENCE

Frontend SWE Intern @ PayPal — San Jose, CA | 2026

- Built and own production Next.js frontend: PM-facing data-viz / hypothesis-testing hub for PayPal's checkout product.
- The hub has already surfaced a significant checkout-friction area- now under active investigation.

Mobile App UI Designer @ tally — Remote | 2026

- Solo designer: created the design system, interaction flows, state logic, and prototypes for a habit-based fintech app.
- Traced complex experiences spanning contracts, onboarding, verification, transaction incentives, and crypto wallets.

Data Visualization & Engineering Intern @ Southern California Edison — Rosemead, CA | 2025

- Designed and optimized ETL pipelines and delivered data solutions to 10+ teams across a utility serving 15M people.

Frontend Web Developer @ Children's Hospital of Philadelphia — Philadelphia, PA | 2025-6

- Developed React and TypeScript features for an ML-driven research-management hub and integrated REST APIs.

ACTIVITIES

Vice President (External), 3x Project Lead, Full-Stack Developer @ Penn Spark | 2024-6

- Lead corporate partnerships and recruiting / career events for Penn's community of creative devs and designers.
- Led redesign and rebuild of Spark's website, improving information architecture, branding, and maintainability.

Governmental Affairs Director @ California Association of Student Councils | 2024

- Authored & passed 3 CA laws, coordinated a 20-organization coalition including ACLU and Human Rights Watch.

SKILLS

Languages: TypeScript, JavaScript, C++, Python, SQL, GLSL/WGSL, Java, HTML/CSS

Web and Data: React, Next.js, Three.js, D3, Node.js, FastAPI, PostgreSQL, Supabase, Pinecone

Graphics and Creative Tech: WebGL, WebGPU, OpenGL, PyTorch, computer vision, Maya, Blender, TouchDesigner

Design and Tools: Figma, Adobe Creative Suite, Docker, AWS, Git/GitHub, Arduino

Achievements: 3x AIME qualifier (top 2.5% nationwide); TOPIK II Level 6